

Notice of Intent to Review Marketplace Solutions

SHOPPING NOTICE: DARPA-SN-26-56

MARKETPLACE: Defense Advanced Research Projects Agency (DARPA) Expedited Research Innovation System (ERIS)

REQUIRING ACTIVITY: DARPA, Information Innovation Office (I2O)

NOTICE DATES: March 12, 2026 – May 31, 2026

1. PURPOSE

Between the dates listed above, DARPA I2O will be actively “shopping” the ERIS Marketplace for innovative, post-competitive, awardable solutions that address the need for new security research tools for artificial intelligence (AI) hardware accelerators (e.g., graphical processing units [GPUs], neural processing units [NPU], and tensor processing units [TPUs]).

2. PROBLEM AND SCOPE

The advancement of AI computing architectures has outpaced the development of security research tools. Commonly available tools such as disassemblers, debuggers, emulators, fuzzers, and interposers used in vulnerability research and reverse engineering are limited or do not exist for analyzing AI hardware accelerators (e.g., GPUs, NPUs, and TPUs). Therefore, the development of these tools is required.

DARPA is exploring the development of tools to conduct security research into AI hardware accelerators. Tools developed may be open-sourced, as appropriate, to encourage the cybersecurity industry to research and improve the security of AI accelerator hardware.

Scope

Vulnerability research tools for AI accelerators are insufficient for in-depth research and analysis of these increasingly important devices. Existing tools are often locked to specific vendors, not available to the public (i.e., proprietary internal vendor tools), or inadequate in their ability to facilitate sophisticated research. Many tools are unable to overcome challenges associated with attempting to debug, emulate, introspect, or reverse engineer AI hardware accelerators (e.g., GPUs, NPUs, and TPUs), and only work on a single thread or process at a time. To enable more sophisticated cybersecurity research, better tools are required.

Research Challenges

Potential awardees will research and develop tools to assist in the vulnerability research and analysis of the AI technology stack (i.e., AI hardware accelerators and software). Key research and innovations are expected to focus on developing tools that advance vulnerability research workflows and tooling for the AI technology stack. Tools are expected to be broadly applicable, and proposals that are limited to an individual system (e.g., vendor make and model) will be viewed unfavorably in their advancement of the state of the art for evaluation purposes.

Parallel Processing

Existing vulnerability research tools for GPUs and other AI accelerators are unable to handle the parallel processing that makes these hardware components valuable. While introspecting individual threads is often sufficient for normal programs, the scale of embarrassingly parallel processors leaves a large blind spot for security researchers. Proposers will present a plan for overcoming challenges stemming from highly parallelized systems and distributed computing architectures resident on AI accelerator hardware.

3. DESIRED CAPABILITIES AND EXPERTISE

While *all submissions will be evaluated against the standard criteria published in the ERIS Master Solicitation*, I2O will be specifically looking for solutions that demonstrate strength in the following areas. These are not new evaluation criteria but rather highlight the specific attributes of interest for this opportunity.

- **Defining the Problem & Current State of the Art:** Strong submissions will demonstrate a deep understanding of the problem space. They should clearly define the scope of the technical challenge they address and compare their proposed solution to the limitations of the current state of the art, particularly regarding the difficulty of analyzing highly parallelized systems.
- **Advancing the State of the Art (Technical Innovation):** The most compelling solutions will be those that offer a significant ability to advance the state of the art in vulnerability research tools for highly parallelized systems, with clear technical insights that address the research challenges in this notice.
- **Team Capability (Expertise and Vision):** Demonstrated, significant expertise in both vulnerability research and parallel processing systems. A strong track record of developing software tools for a broad community (e.g., highly adopted open-source software contributions). Direct experience developing vulnerability research tools is advantageous but not required.
- **Impact (Defense and Commercial Use Care):** Strong submissions will advance AI accelerator hardware security by developing and open-sourcing new tools not yet in existence, or readily available, to the Defense and Commercial industries, for vulnerability research and analysis of the AI technology stack.

4. HOW TO RESPOND

This notice is a call for solution to be flagged for review.

For Existing Awardable Solutions:

If you have a solution already in the ERIS Marketplace that aligns with this need, send an email to ERIS@darpa.mil with:

- Subject Line: Shopping Notice Response: DARPA-SN-26-56 - [Your Company Name]
- Body: The Title and Video URL of your existing awardable solution.

For New Submissions:

If you have a new solution, please submit it to the "Strategic Systems" Topic Area, listed below, via the ERIS portal.

Developing advanced technologies for improved resilience, efficiency, and effectiveness of strategic systems. This topic area supports technologies for critical infrastructure and military command-and-control systems (networks and applications across strategic, command, operational, and tactical edges).

To ensure it is associated with this notice, please use the following format for your video title:

- Video Title Format: DARPA-SN-26-56: [Your Solution Title]
(Example: DARPA-SN-26-56: AI Accelerator Analysis Tool)

After submitting, please also send a notification email to ERIS@darpa.mil with:

- Subject Line: Shopping Notice Response: DARPA-SN-26-56 - [Your Company Name]
- Body: The Title of your new submission and its Date of Submission.

Solutions submitted during this window that are deemed "awardable" through the standard ERIS assessment process will be included for consideration by I2O.

5. RESOURCES

- ERIS Master Solicitation: [SAM.gov link](#)
- To submit a new solution: [DARPAConnect ERIS Portal](#)